

Final Project

Final Project: Mini Search Engine

Build a small search engine that indexes a folder of files and lets the user search through them with the query types we covered. Users should be able to:

- Pick which file formats to include in the index.
- Type any search query (boolean, phrase, fuzzy, wildcard).
- Filter by a date range (modification date) and by file type.
- See highlighted snippets in the results, with paging.
- Get a "Did you mean?" suggestion when nothing matches.
- See basic stats about what's in the index.

How you build it is your choice.

File Formats Required

Your engine must extract searchable text from:

- TXT
- PDF
- JSON
- CSV
- Excel (.xlsx)

The user must pick which formats to index before building. For example, "only index PDFs and TXTs from this folder." How they pick (checkboxes, command-line flags, a config file) is up to you.

For JSON, CSV and Excel, you decide what counts as "the document's text". The whole file as one document, each row or object as a separate document, all string values flattened, etc. Document your choice.

Search Features Required

Every feature below must work in your search box. The user shouldn't need a separate button for each. `QueryParser` handles most of them automatically.

- Boolean operators: `AND` , `OR` , `NOT` , grouping with `()`
- Phrase queries: `"information retrieval"`
- Fuzzy queries: `retrival~` , `retrival~2`
- Wildcard queries: `inform*` , `?earch`
- Date range filter on the file's modification date (read it from the file with `os.path.getmtime`)
- File-type filter at search time
- Highlighted snippets in results
- "Did you mean?" on zero results
- Paging: show 5 results per page with next/previous navigation

Stats the User Should See

At any time, the user should be able to see:

- Total number of indexed documents.
 - Breakdown by file type (pdf: 4, txt: 8, ...).
 - Top 10 most frequent terms in the indexed content.
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UI Expectations

You can build a CLI, a desktop GUI, or a web app. Whatever you pick, the interface must let the user do all of these:

1. Pick file formats before building the index.
2. Trigger an index build / rebuild.
3. See a summary after the build (how many files, by type).
4. Type a search query in any supported syntax.
5. Set a date range filter on the modification date (optional per search).
6. Set a file-type filter at search time (optional per search).
7. See results with filename, file type, score, modification date, and highlighted snippet.
8. See a "Did you mean?" suggestion when a search returns nothing.
9. Move between result pages (5 per page).
10. Open the stats view.

If your UI doesn't allow one of these actions, that feature is missing.